ZAIN BUGHIO

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EDUCATION

University of Waterloo, B.Math - Double Major in Applied Math (Scientific ML) & Statistics — Expected 2029

SKILLS

Programming Languages	Java, Python, JavaScript, SQL, C, TypeScript
Frameworks and Libraries	React, PyTorch, FastAPI, ONNX, MLflow, Tailwind CSS, Streamlit, Three.js, GSAP
Tools and Platforms	GitHub, Jira, PowerShell, Linux, VS Code, Vercel, Tableau, Supabase, Blender

EXPERIENCE

Network Engineering Intern — May 2026 - August 2026
Waterloo Regional Health Network — *Waterloo, ON*

- Managed **IPv4-based laptop imaging** and bulk software deployment via institutional image servers, scripting installs with **PowerShell and Bash** to migrate **100+ staff workstations** for a hospital-wide **Windows 10 → 11** rollout.
- Spearheaded a full-scale **live hospital data center remediation**, tracing, tagging, and decommissioning **500+ lbs of RJ45/Ethernet and fiber optic cabling**, auditing and unplugging dead connections from **Cerner servers, Fortinet firewalls, and core routers** across storage arrays holding up to **3 petabytes** of clinical data, all with **zero downtime**.
- Worked with senior security analysts on live **Fortinet firewall, Cisco DNS filtering, Bell Cyber, and CrowdStrike SIEM** operations, mapping the traffic path (**endpoint → DNS blocker → switch → firewall → router → internet**) and comparing **Corp vs. Guest WiFi** firewall segmentation and Windows Event Viewer telemetry routing.

AI Engineer — July 2026 - Present
Rubricle (AI Startup) — *Remote*

- Engineered a **layered prompt architecture** spanning persona, task, and audit layers for an autonomous LLM agent built on **Hermes Agent**, optimizing instruction design for personalization and consistent cold-outreach tone.
- Architected and deployed a **Dockerized AI agent pipeline**, integrating **Hermes Agent** with **ScrapeGraph** for automated web-based data enrichment and **n8n** for workflow orchestration, while configuring persistent volumes.
- Deployed a **multi-profile Hermes gateway** with isolated profile routing and **profile-specific authentication**, restricting port exposure to the n8n Docker gateway while enabling /p/<profile>/ API access through a single port **8642**.

Software Engineer — May 2025 - December 2025
PixelsBoost — *Milton, ON*

- Engineered **6 production-grade full-stack websites** using **React, HTML/CSS, and JavaScript**, integrating **3 mission-critical third-party APIs (Stripe, Google Maps, SendGrid)** to enable **$10K+ USD** in monthly payment processing, collectively serving **1,500+ monthly active users** with **99%** uptime.
- Spearheaded delivery of **5 simultaneous client projects** in an **agile environment**, authoring **200+ commits** across **20+ feature branches** while mentoring 2 designers and resolving **15+ merge conflicts**.
- Drove a **40%** performance uplift across client websites via WebP compression, lazy loading, and **Cloudflare CDN**, elevating Google Lighthouse scores from 65 → 86 and cutting bounce rates by **15%**.

PROJECTS

Frame | TypeScript, Python, Electron, MLX, LoRA, GGUF, RAG, FAISS

- Built a fully **local AI coding IDE** on a **VS Code fork** that assembles context from a **multi-tier memory system**, **RAG index**, and **workspace knowledge graph**, then routes requests to a locally-hosted **Qwen2.5-Coder-7B** model running on-device via **llama.cpp**, keeping all code and data private with **zero cloud dependencies**.
- Built a **personalization pipeline** that captures coding behavior (accepts, edits, reverts) through an **observation service**, validates patterns with a **preference analyzer**, and confirms them via a **human-in-the-loop approval UI** before training, then fine-tunes **LoRA adapters** using **MLX**, fusing and quantizing them into **GGUF** format for offline inference.

Wanderers (Hack Canada) | Next.js, Supabase, Gemini, Tailwind CSS, FastAPI, Scikit-learn, Cloudinary, Vercel

- Campus social app with live activity "bubbles," and a **Gemini + K-Means recommendation engine (72.2% Hit Rate@6)**.

CoopTrack | React, Vite, Tailwind CSS, Supabase, Google Gemini, Vercel

- Application tracker that uses **Gemini** to **parse** job postings into a structured, term-based pipeline (Applied → Offered).

Strata | Rust, Zstd, LZ4, Parquet

- **lossless engine** storing 179M Nasdaq records **10% smaller than Parquet/Zstd** at **4.54M records/s, 2.6× faster**.